256 Translation Accuracy and Character Variety in Fast Input Method for Mobile Terminal

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1 Introduction

Recently, performance of a mobile terminal such as a mobile phone is greatly increasing. The mobility of a mobile phone is very good because the typical size is small. However, it has only number keys, which are 0, 1, ..., 9,* and #, because of its own size. To input characters into a mobile phone, the letter cycling input method is most commonly used. The inputted character is decided by the chosen key and the number of pressing it. For example, a user chooses the key "2" and presses three times in order to input the alphabet "c". Since this input method needs several keystrokes per one character, it is troublesome for users. Therefore, the method is demanded which enable to promptly and easily input characters.

We have proposed the input methods on a mobile terminal for Japanese[1] and for Chinese[2]. The former method translates the number characters into the Japanese sentence and is expressed by N-JT. The latter one translates the number characters into the Chinese sentence and is expressed by N-CT in this paper. A user is able to input characters rapidly and easily in our proposed method. For example, a user presses the key "2" only one time in order to input "c". However, it means "a" and "b" also. Therefore, the inputted number string is ambiguous. However, the system is able to disambiguate the number string by Inductive Learning[3] and translate into the correct sentence.

This paper shows the outline of our proposed method and the relation between the translation accuracy and the variety of characters in our proposed method.

2 Outline of Our Proposed Method

A user inputs the number string corresponding to the pronunciation of the intended sentence in our proposed method. Some number characters are assigned to each key on a mobile phone and the system needs one keystroke per one character. Although a user is able to input rapidly and easily, the inputted number string is ambiguous because the string corresponds to not only the intended sentence but also many sentences. However, the system disambiguates the number string by Inductive Learning. The system automatically acquires segments from the sentence using Inductive Learning. Then, the system is able to adapt to the target field dynamically even if the target field changes. Therefore, the system based on our proposed method is able to translate the number string into the sentence intended by the user.

3 Evaluation Experiment

We evaluate the translation accuracy and the variety of characters on N-JT and N-CT.

3.1 Data and Procedure

The data has three sets which are Unix Online Manual in Japanese, the first author's E-mail in Japanese and Sinica Corpus in Chinese. Each set has 50,000 characters. The variety for characters of each set is evaluated by the entropy. It is expressed by H and defined as:

$$H(S) = -\sum_i P_i \log_2 P_i \quad (1)$$

where S is a set of characters and P_i is a probability for a character of the set. The sets of "Unix" and "Mail" are translated by N-JT and the set of "Sinica" is translated by N-CT. The translation accuracy is evaluated by *F-measure* and defined as:

$$F\text{-}measure = \frac{(1+b^2)\cdot precision \cdot recall}{b^2 \cdot precision + recall} \quad (2)$$

where b is a coefficient. The value of b is 1 in this experiment.

Table 1 Entropy and F-measure.

	Entropy[bit]	F-measure[%]
Unix(in Japanese)	7.22	71.8
Mail(in Japanese)	7.53	63.8
Sinica(in Chinese)	8.47	50.4

3.2 Results and Considerations

Table 1 shows the entropy and F-measure for each set. The entropy for Chinese is higher than that for Japanese. The entropy means variety for characters of a set. Therefore, the F-measure of the Chinese set is lower than that of the Japanese sets. The kind of Chinese characters is more than that of Japanese characters. Then, the entropy is high and the F-measure is low for Chinese. We consider that the variety of characters is low when the target field is limited. To improve the translation accuracy, the system needs increasing its own adaptability to the target field.

4 Conclusion

This paper shows the outline of our proposed method and the relation between the translation accuracy and the variety of characters in our proposed method. In the experiment, it was confirmed that the translation accuracy for the set of which the entropy is high is low. A future work is to increase the adaptability of our proposed method to the target field.

References

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